5 In an effort to address pollution caused by industrial nitroaromatics and agricultural nitrates, researchers have developed a dual-function electrochemical system that converts these nitrogen-containing wastes into useful products. At the cathode, the electrolytic reduction of nitrobenzene to phenylamine is described in Fig. 5.1.

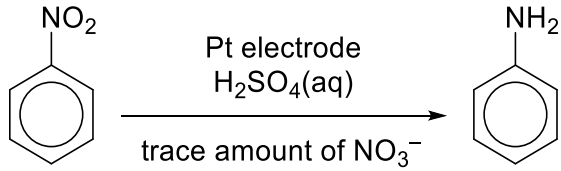

Fig 5.1

(a) State the reagents and conditions for the conversion of nitrobenzene to phenylamine in a laboratory. [1]

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(b) **(i)** Given the nitrogen in nitrobenzene has an oxidation state of +3, describe the change in oxidation state of the nitrogen, in Fig. 5.1. [1]

(ii) Hence or otherwise, write the half-equation for this reaction. [1]

(iii) During a 4-hour electrolysis, a steady current of 2 A was passed. However, only 2.79 g of phenylamine was formed.

Using your answer to **(b)(ii)**, calculate the theoretical amount of electrons required to form 2.79 g of phenylamine. [1]

(iv) Faradaic efficiency describes the efficiency with which charge is transferred in an electrolysis system and is given by the equation below.

$$\text{Faradaic efficiency} = \frac{\text{charge required}}{\text{charge passed}} \times 100\%$$

Hence, calculate the Faradaic efficiency of the electrolysis in **(b)(iii)**. [3]

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(c) The use of the *Data Booklet* is relevant to this question.

(i) A student suggests that the trace amounts of NO_3^- ions at the cathode could also be reduced to NH_4^+ ions. Discuss how electrode potential and concentration might influence this competition.
(The $E^{\ominus}$(nitrobenzene | phenylamine) is +0.79 V.) [3]

(ii) Suggest other possible side products at the cathode. [1]

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(d) Rank the following compounds in order of increasing basicity and explain your reasoning.

phenylamine, methylamine, ethanamide [4]

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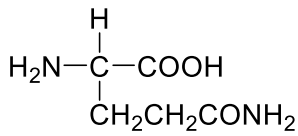
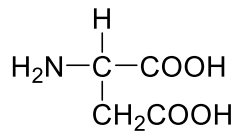
(e) The structures of two amino acids, glutamine and aspartic acid are given below.



glutamine



aspartic acid

(i) Suggest a simple chemical test to distinguish between glutamine and aspartic acid. [2]

(ii) The three pK_a values associated with aspartic acid are 1.99, 3.90 and 9.90.

Draw the structures of the predominant species of aspartic acid at

- pH 3
- pH 8
- pH 11 [3]

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[Total: 20]